

Run report

2026-09-10T132409Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	2,113,982,479	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	3.705×10^{-4}	best_epoch	38
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	58.074	81.1	train
transfer	1.449		train
compile	6.796		train
epochs	48.059		train
select	1.500		train
particle_mmd	0.556	0.8	train
save	0.071	0.1	train
data	1.587	2.2	cache hit – load
load	1.526	2.1	load
plots	9.781	13.7	load
evaluate	0.008	0.0	load
total	71.604	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	208.9	138.8	33.6	6.729	96.8	11.57	94.5
$\ln \rho$	227.9	20.71	90.9	12.61	94.5	13.22	94.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	262.6	12.84	95.1	8.776	96.7	11.61	95.6
w	265.6	9.978	96.2	7.845	97.0	11.41	95.7
λ_2^1	255.4	9.215	96.4	8.637	96.6	13.10	94.9
Splitting and 2-prong substructure							
$z_g^\dagger$	60.45	60.45	0.0	13.60	77.5	13.68	77.4
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	30.64	74.8	17.68	85.5
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	224.2	177.3	20.9	6.087	97.3	21.68	90.3
n_{ch}	220.8	172.3	22.0	6.443	97.1	22.87	89.6
f_{ch}	52.86	40.94	22.6	4.406	91.7	7.417	86.0
p_T^D	256.8	30.80	88.0	3.274	98.7	18.57	92.8
q	79.67	17.51	78.0	3.829	95.2	7.496	90.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	6.762	3.666	45.8	0.07682	98.9	0.09512	98.6
$\ln \rho$	8.383	0.4017	95.2	0.1644	98.0	0.1516	98.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1299	98.6	0.1209	98.7
w	9.139	0.4527	95.0	0.1223	98.7	0.1261	98.6
λ_2^1	7.776	0.1728	97.8	0.1307	98.3	0.1294	98.3
Splitting and 2-prong substructure							
$z_g^\dagger$	0.6723	0.6723	0.0	0.09993	85.1	0.1035	84.6
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.2792	91.7	0.1638	95.1
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	9.640	7.633	20.8	0.1202	98.8	0.1830	98.1
n_{ch}	7.512	6.314	16.0	0.1042	98.6	0.1431	98.1
f_{ch}	1.033	0.7960	22.9	0.08796	91.5	0.09904	90.4
p_T^D	8.223	1.311	84.1	0.07408	99.1	0.1520	98.2
q	1.576	0.2512	84.1	0.07331	95.3	0.08472	94.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

VLC divergence [$\times 10^3$]

[illegible]

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	111.7	94.08	15.8	104.1	6.8	89.80	19.6
$\ln \rho$	253.1	32.57	87.1	12.81	94.9	14.25	94.4
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	303.0	18.71	93.8	14.40	95.2	8.400	97.2
w	284.1	12.29	95.7	15.10	94.7	6.554	97.7
λ_2^1	253.8	11.13	95.6	18.51	92.7	6.227	97.5
Splitting and 2-prong substructure							
$z_g^\dagger$	113.4	113.4	0.0	23.95	78.9	15.12	86.7
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	34.77	86.6	10.42	96.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	256.8	204.8	20.2	43.59	83.0	20.68	91.9
n_{ch}	234.6	188.6	19.6	43.39	81.5	22.69	90.3
f_{ch}	45.78	45.33	1.0	4.707	89.7	9.442	79.4
p_T^D	236.3	55.36	76.6	25.32	89.3	16.85	92.9
q	79.37	18.15	77.1	5.345	93.3	7.897	90.1

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JS divergence [$\times 10^3$]

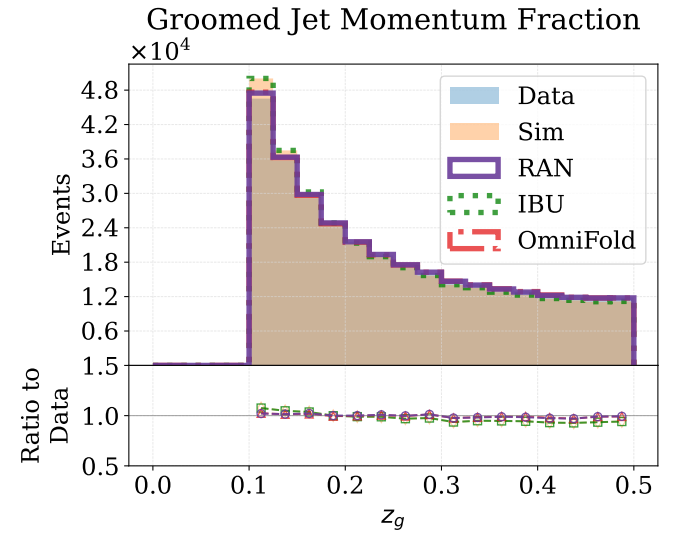
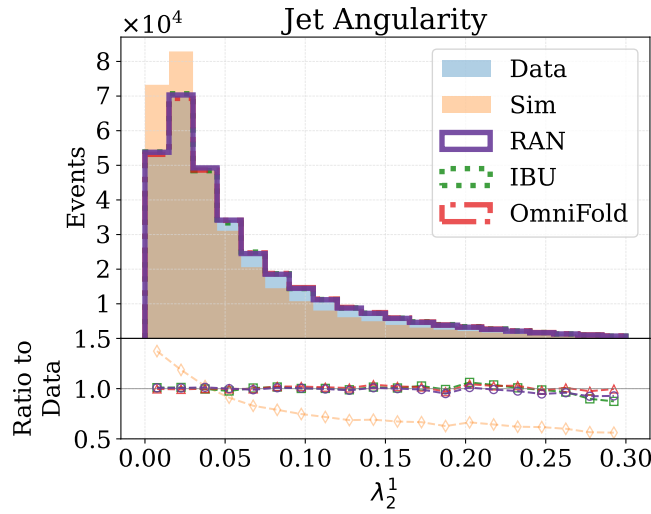
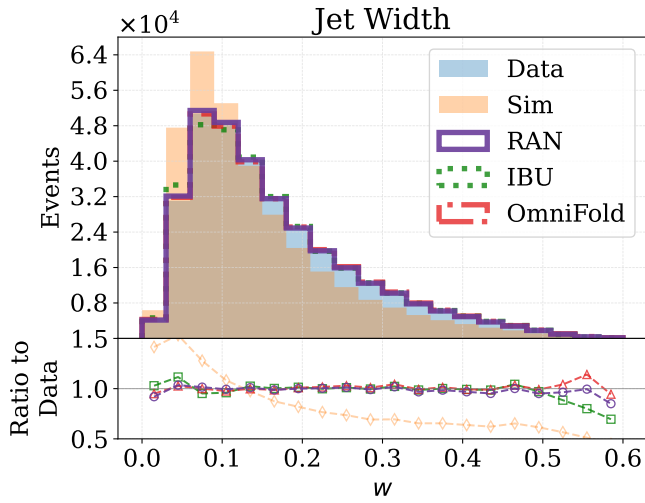
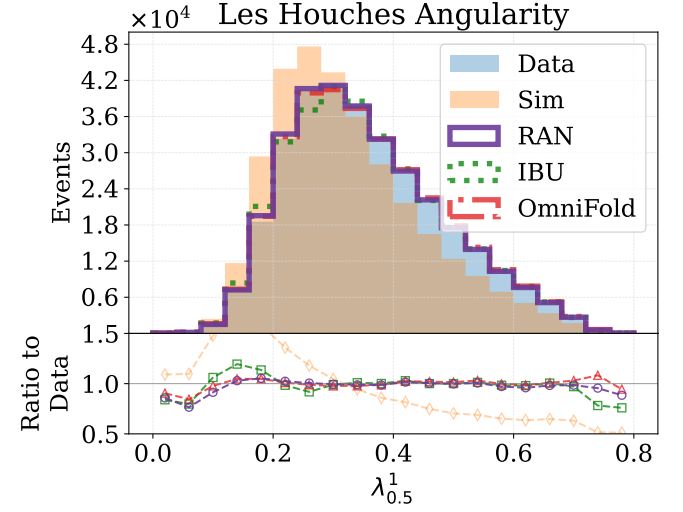
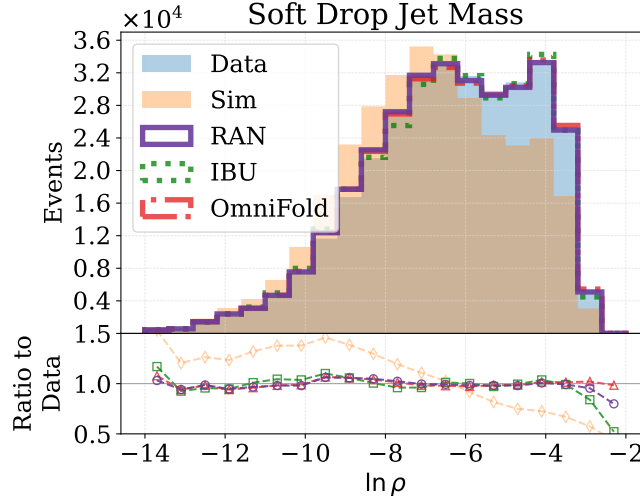
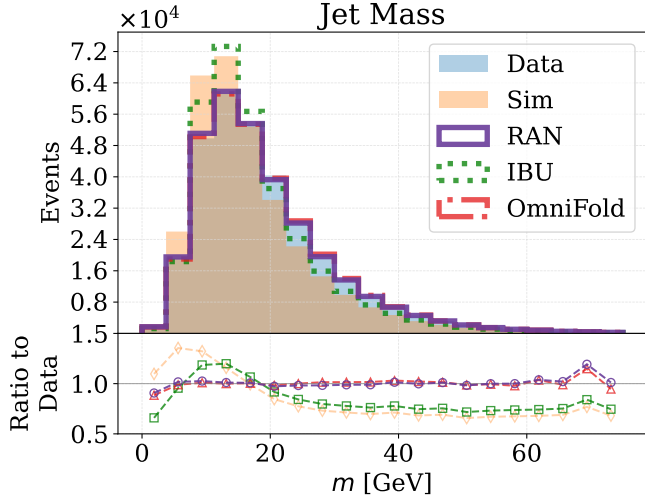
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	1.963	11.86	−504.3	2.045	−4.2	1.424	27.5
$\ln \rho$	9.878	1.731	82.5	0.2625	97.3	0.1730	98.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.1865	98.5	0.1850	98.5
w	11.39	1.465	87.1	0.2037	98.2	0.1999	98.2
λ_2^1	8.824	0.8335	90.6	0.1872	97.9	0.1384	98.4
Splitting and 2-prong substructure							
$z_g^\dagger$	2.041	2.041	0.0	0.2143	89.5	0.1317	93.6
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.4798	95.4	0.1377	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	7.842	12.83	−63.7	0.3742	95.2	0.2267	97.1
n_{ch}	6.447	9.773	−51.6	0.3046	95.3	0.1707	97.4
f_{ch}	1.025	0.9435	7.9	0.09836	90.4	0.1305	87.3
p_T^D	8.342	4.029	51.7	0.4205	95.0	0.2556	96.9
q	2.027	0.9958	50.9	0.1058	94.8	0.1020	95.0

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

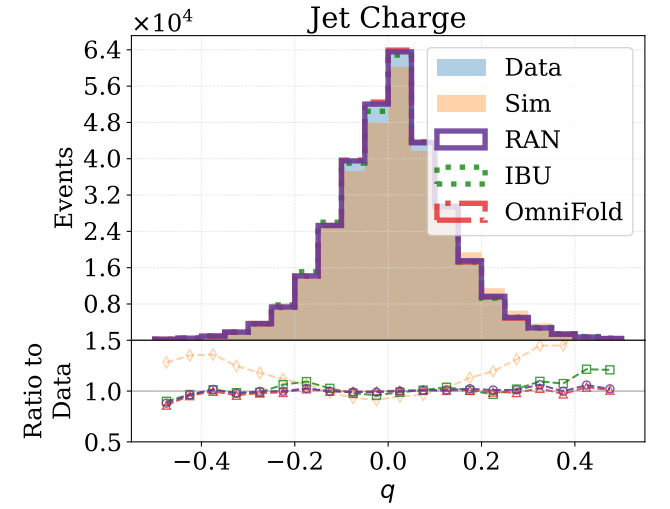
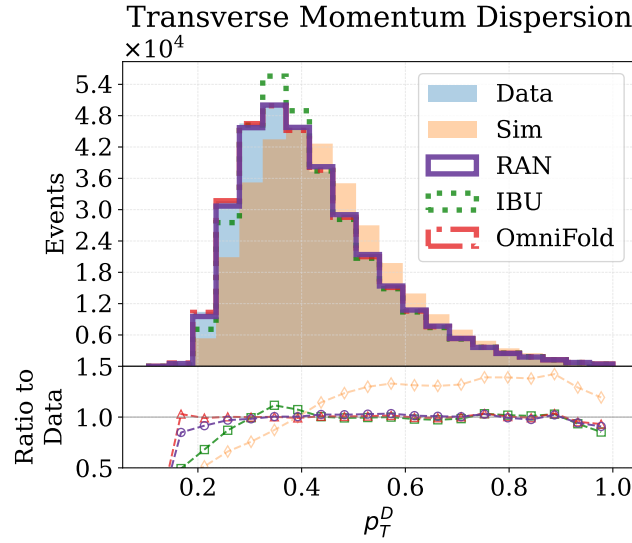
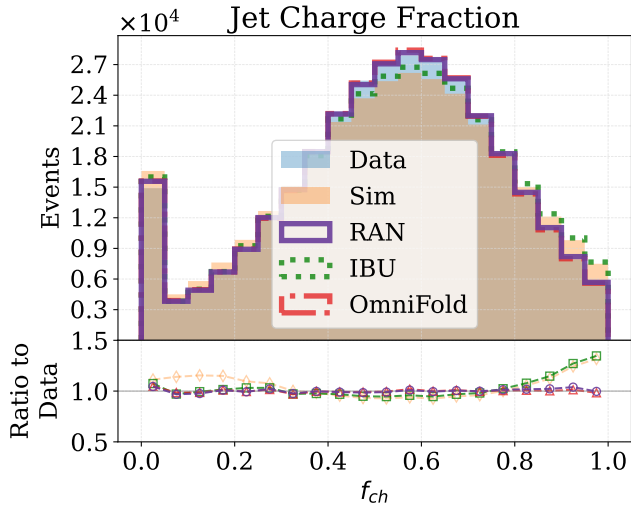
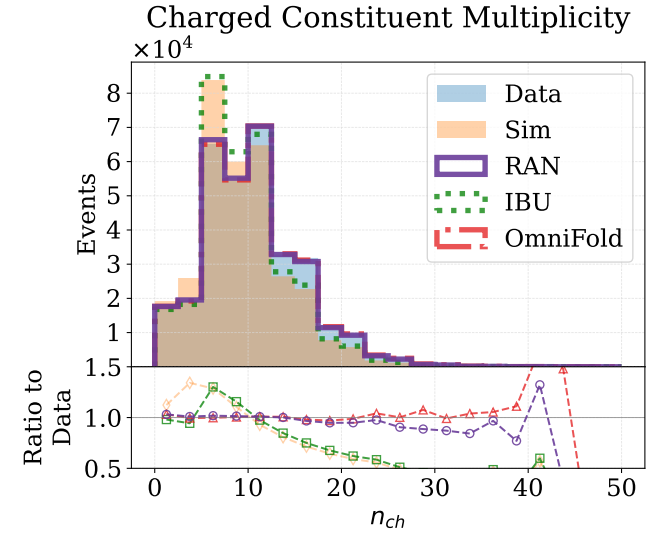
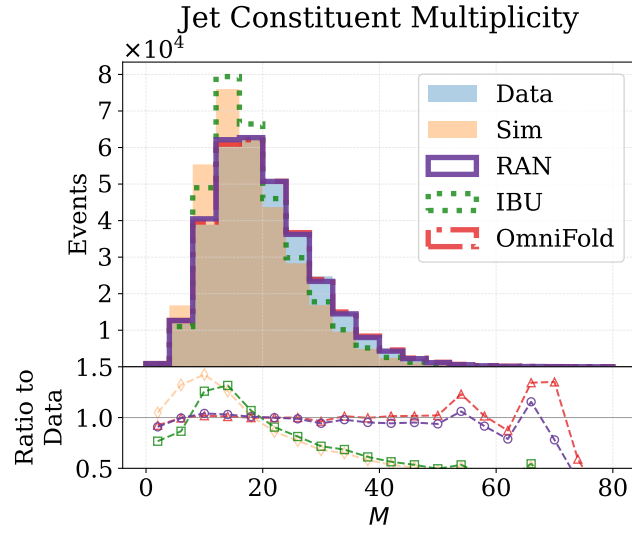
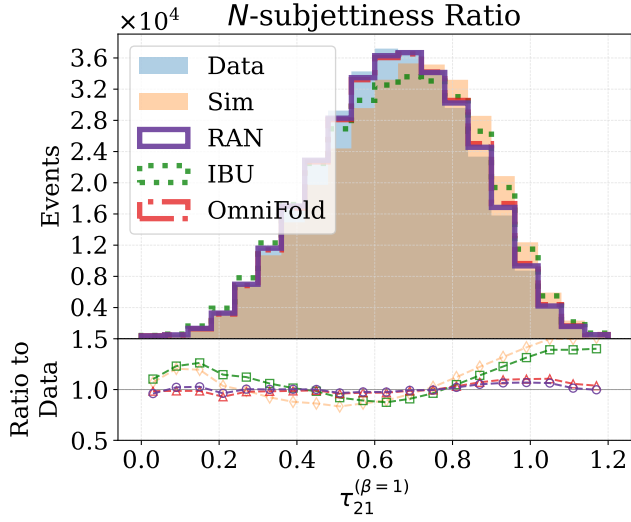
VLC divergence [$\times 10^3$]

[illegible]

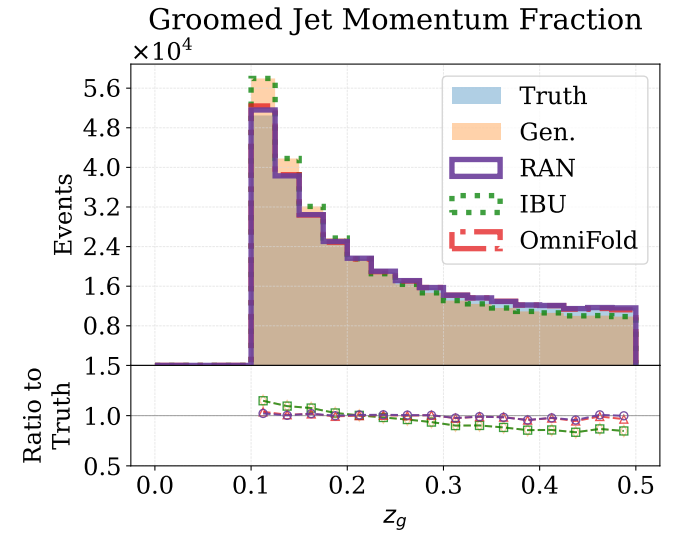
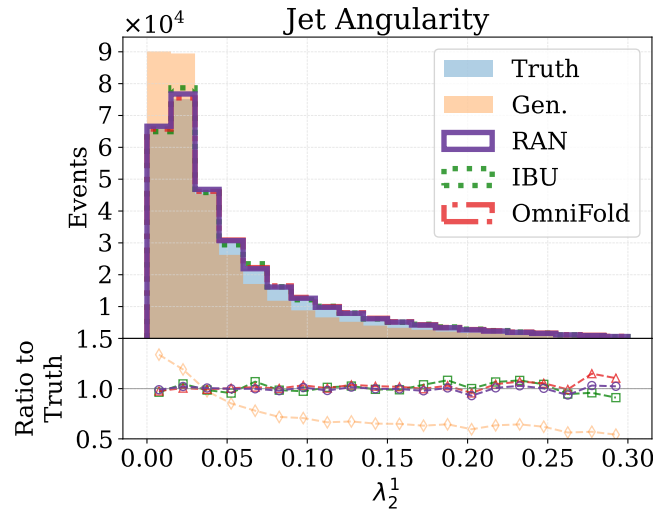
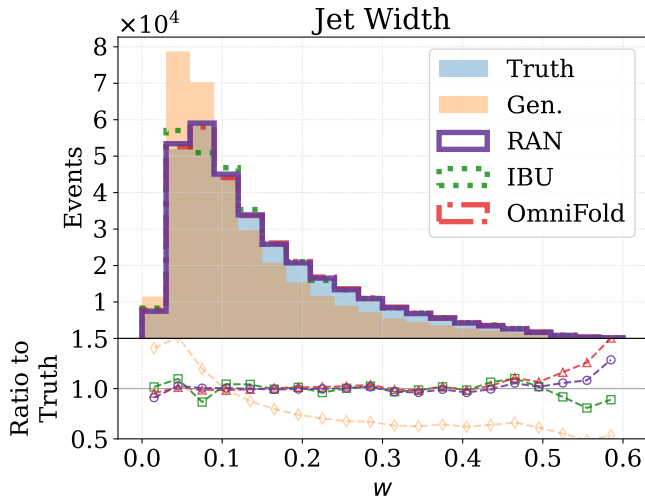
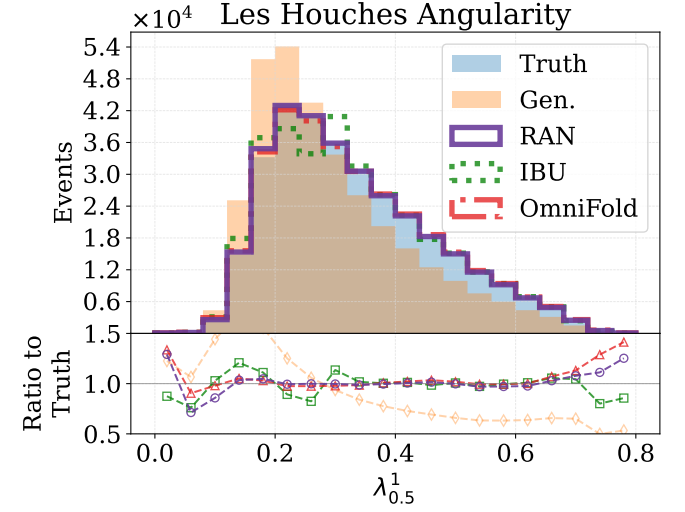
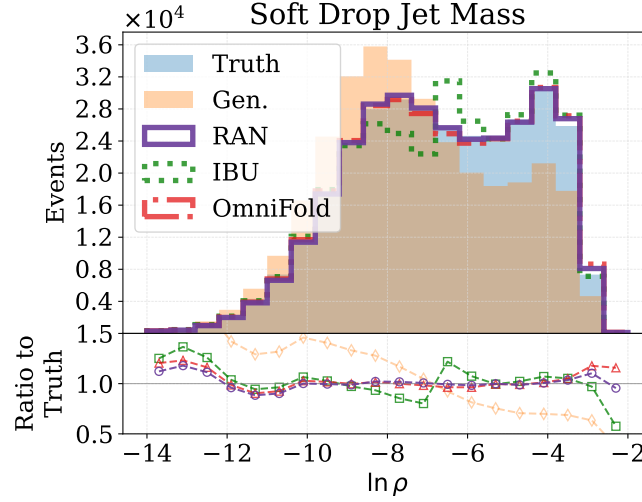
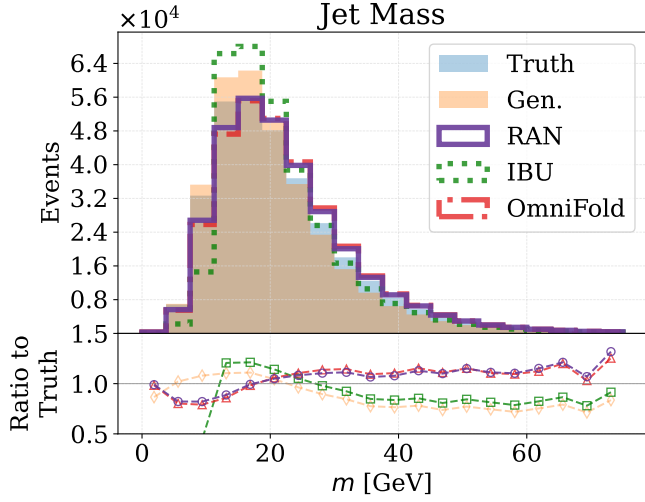
Detector Level (1 of 2)



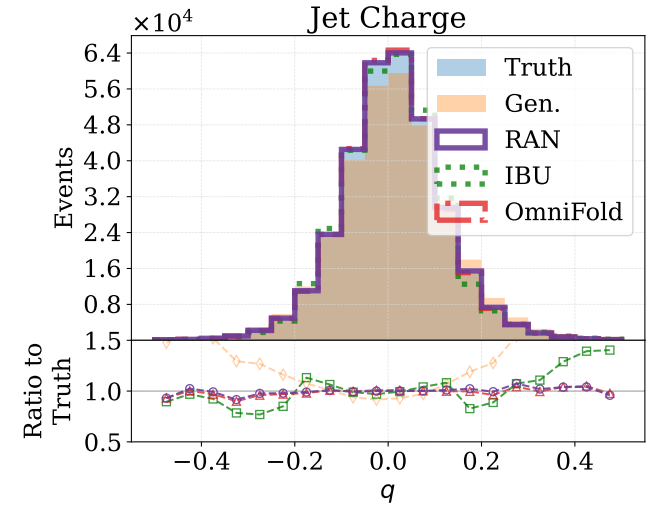
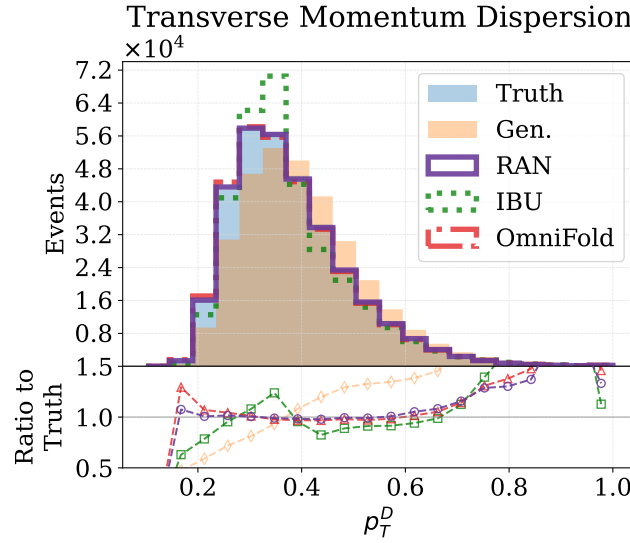
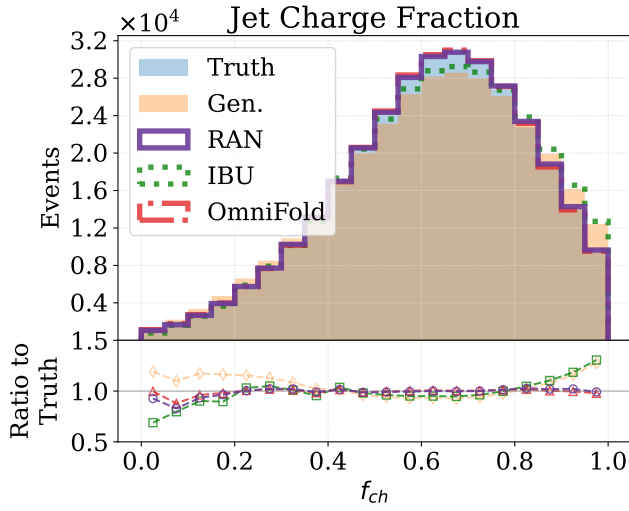
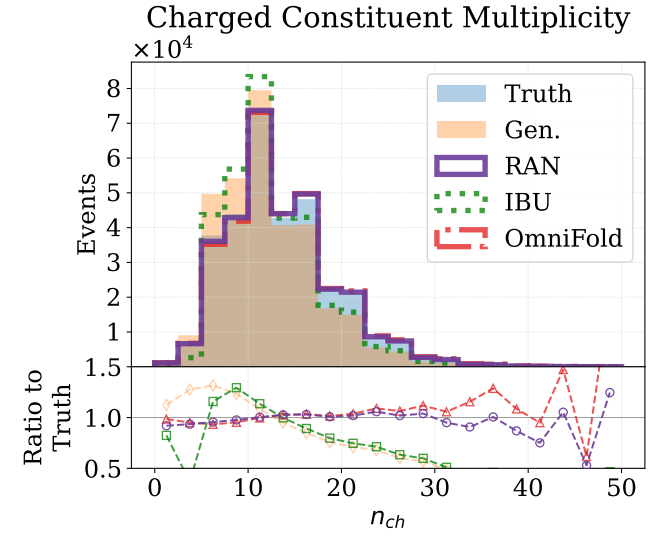
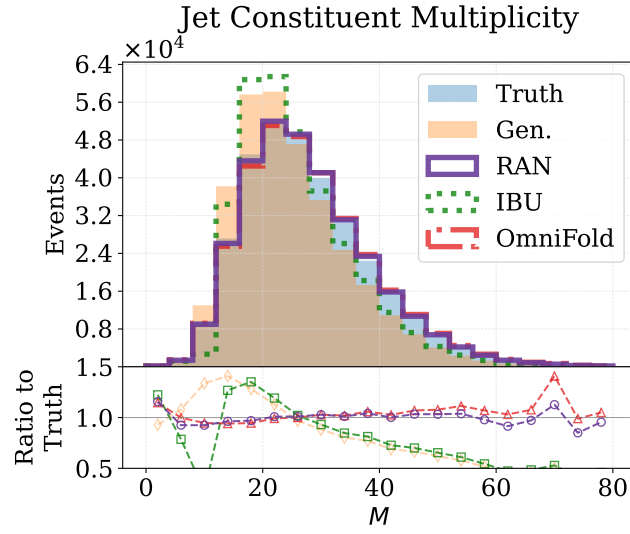
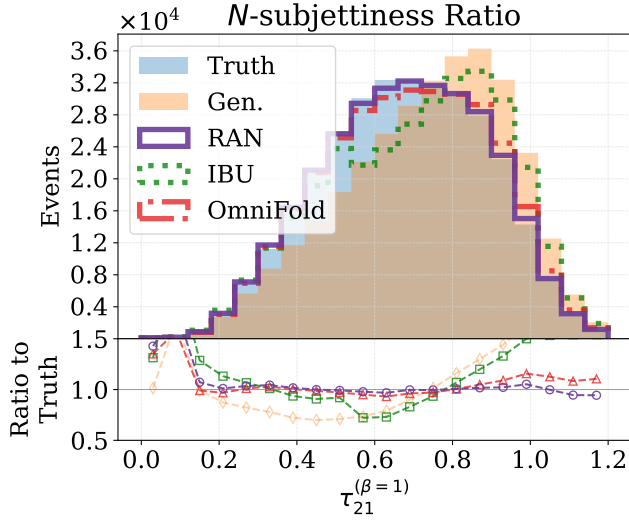
Detector Level (2 of 2)



Particle Level (1 of 2)

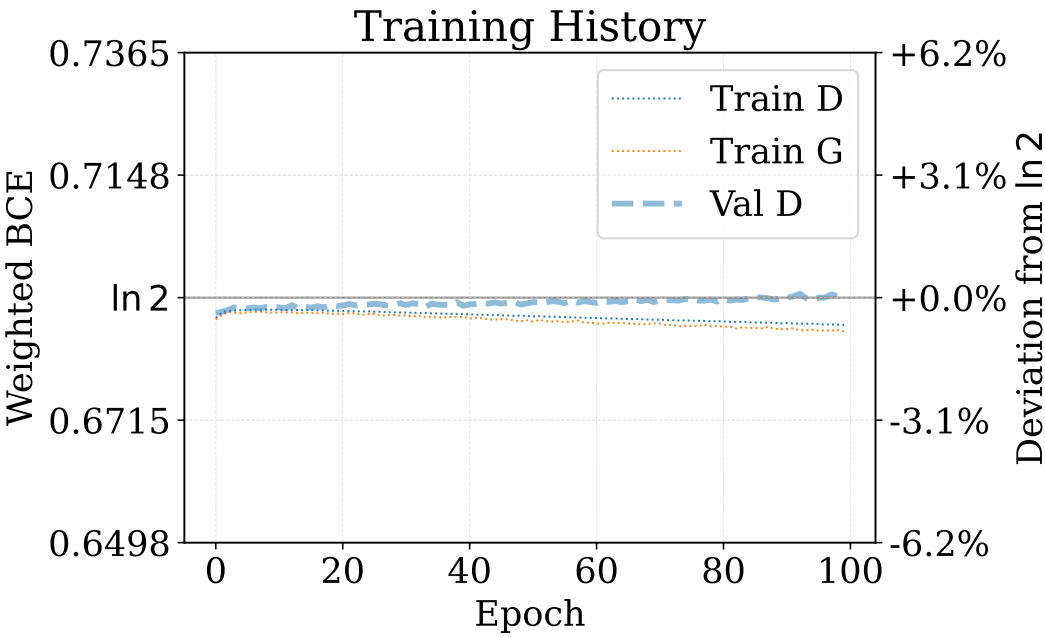


Particle Level (2 of 2)



Training diagnostics

Training losses



Selection criteria

